

Partial Fractions



Distinct linear factors

- 1 Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.
 - 2 Express $(3x - 1)/(x(x - 3))$ in partial fractions.
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Repeated linear factors

- 3 Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.
 - 4 Express $(2x^2 - 3x + 1)/(x(x - 1)^2)$ in partial fractions of the form $A/x + B/(x - 1) + C/(x - 1)^2$.
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Irreducible quadratic factors

- 5 Express $(3x^2 + 2)/((x - 1)(x^2 + 1))$ in partial fractions of the form $A/(x - 1) + (Bx + C)/(x^2 + 1)$.
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Applying partial fractions to integration

- 6 Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2)) dx$, using the decomposition found earlier in this worksheet.
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Applying partial fractions to a binomial series expansion

- 7 Express $1/((1 - x)(1 - 2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x .
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Combining terms before decomposing

- 8 Express $(7 - x)/((x + 1)(x - 2))$ in partial fractions.
 - 9 Express $(x^2 + x + 1)/(x(x + 1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)
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Applying partial fractions to a rate problem

- 10 A population model gives the rate of change $dP/dt = 1/(P(10 - P))$. Express $1/(P(10 - P))$ in partial fractions of the form $A/P + B/(10 - P)$.
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Partial fractions with three distinct linear factors

- 11 Express $(6x^2 - 2x - 1)/((x - 1)(x + 1)(x - 2))$ in partial fractions.
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Partial fractions with an improper fraction and repeated factor

- 12 Express $x^3/(x - 1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.
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Using partial fractions to sum a series

- 13 Express $1/(n(n + 1))$ in partial fractions, and use the result to find $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.
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Decomposing a fraction with a repeated quadratic-adjacent structure

- 14 Express $(x + 5)/((x - 1)(x + 1)^2)$ in partial fractions of the form $A/(x - 1) + B/(x + 1) + C/(x + 1)^2$.
- 15 Express $(2x^2 + x - 1)/(x^2(x + 1))$ in partial fractions of the form $A/x + B/(x^2) + C/(x + 1)$.
- 16 Use partial fractions to find the Maclaurin (power series) expansion of $1/((1 - x)(2 - x))$ up to the term in x^2 , first decomposing into the form $A/(1 - x) + B/(2 - x)$.
- 17 Express $(4x^2 - 5x - 15)/((x + 1)(x - 2)(x + 3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).
- 18 A chemical reaction's rate is modeled by $dC/dt = 1/((C + 2)(C - 1))$, where C is a concentration. Express $1/((C + 2)(C - 1))$ in partial fractions of the form $A/(C + 2) + B/(C - 1)$, and then use the result to write the general antiderivative $\int 1/((C + 2)(C - 1)) dC$ in terms of natural logarithms.
- 19 Use the decomposition $1/(3x) + 8/(3(x - 3))$ found earlier in this worksheet for $(3x - 1)/(x(x - 3))$ to evaluate $\int_4^5 (3x - 1)/(x(x - 3)) dx$, giving your answer as a single logarithm expression.

- 20 This question has three parts. (a) Factor $x^2 - 4$ and hence express $(x + 8)/(x^2 - 4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x + 8)/(x^2 - 4) dx$. (c) Hence evaluate $\int_3^5 (x + 8)/(x^2 - 4) dx$, giving your answer as a single expression involving natural logarithms.
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Combining partial fractions back into a single fraction

- 21 Combine $2/(x + 1) + 3/(x + 2)$ into a single fraction.
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Identifying the correct form of a decomposition

- 22 Which of the following is the correct form for the partial fraction decomposition of $(3x + 1)/((x - 2)^2(x + 1))$?
- a. $A/(x-2)+B/(x-2)^2+C/(x+1)$
 - b. $A/(x-2)+B/(x+1)$
 - c. $A/((x-2)^2)+B/(x+1)$
 - d. $(Ax+B)/(x-2)^2+C/(x+1)$
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Solving a separable differential equation using partial fractions

- 23 Solve the differential equation $dP/dt = P(10 - P)/100$, given that $P = 5$ when $t = 0$. Use the partial fraction decomposition $1/(P(10 - P)) = 1/(10P) + 1/(10(10 - P))$ found earlier in this worksheet to integrate, and give P explicitly as a function of t .
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Explaining when polynomial division is required

- 24 Explain why polynomial division must be performed before applying partial fraction decomposition to $x^3/(x - 1)^2$, but not to $(3x + 1)/(x - 1)^2$.
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Repeated irreducible quadratic factor

- 25 Express $(2x^3 + 3x^2 + 4x + 1)/(x^2 + 1)^2$ in partial fractions of the form $(Ax + B)/(x^2 + 1) + (Cx + D)/(x^2 + 1)^2$.
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Verifying a decomposition by substitution

- 26** Verify the decomposition $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$ found earlier in this worksheet by substituting $x = 0$ into both sides.
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Choosing efficient substitution values

- 27** When decomposing $(2x + 3)/((x - 4)(x + 5))$ using the equation $2x + 3 = A(x + 5) + B(x - 4)$, which two values of x should be substituted to find A and B most efficiently, and why?